

Qatar-4b

R-2553

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_210110 · created 2026-08-26T01:05:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459224.6926 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.02
Transit depth (Rp/Rs)^2	1.57 +/- 0.5 [%]
Orbital Inclination (inc)	86.39 +/- 1.52
Airmass coefficient 1 (a1)	0.97 +/- 0.017
Airmass coefficient 2 (a2)	0.0185 +/- 0.0039
Scatter in the residuals of the lightcurve fit is	1.8 %
Best Comparison Star	#2 - [129, 164]
Optimal Method	PSF photometry
Transit Duration (day)	0.0821 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.125 ± 0.02 vs 0.138 (deviation 0.65σ)
Duration fit vs published	0.0821 d vs 0.0892 d (deviation 0.89σ)
Mid-time O-C	3.6 min
Depth SNR	6.2
Residual scatter	1.8 %

Light curve

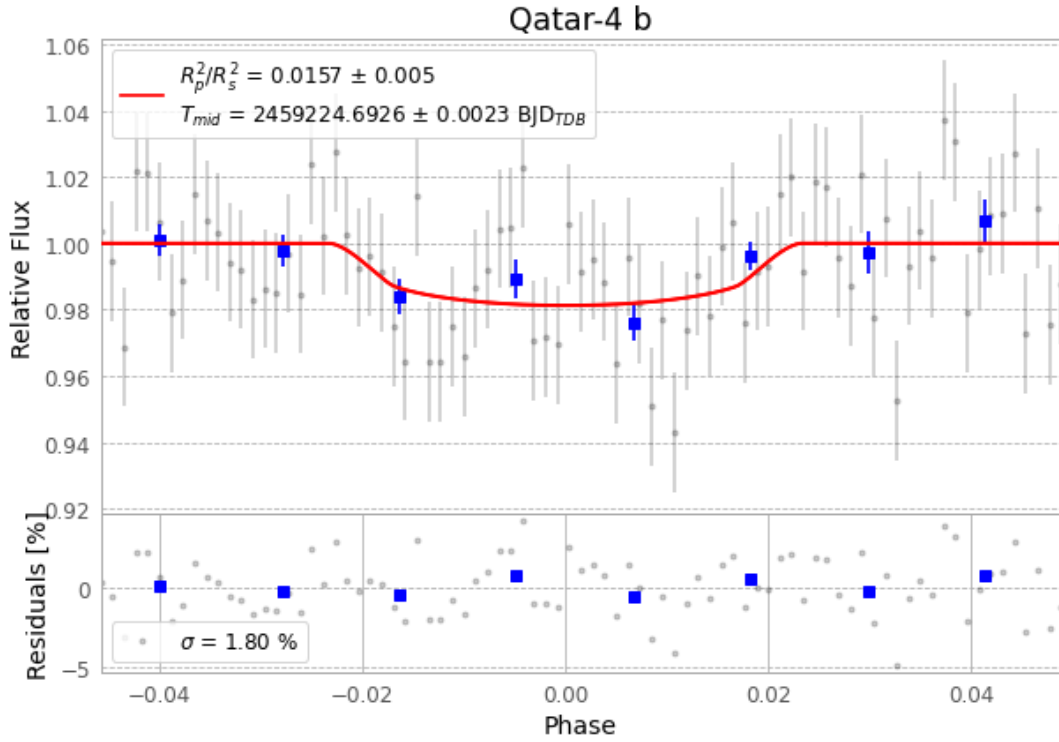


Figure 1 — FinalLightCurve_Qatar-4 b_2021-01-09.png (SHA-256 c16658f10058ce43...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [440, 300], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1b097c4025ec9244...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

83 FITS frames (55 MB), Qatar-4210110023612.FITS → Qatar-4210110064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-210110.log (1a7349480fa0...), FinalLightCurve_Qatar-4 b_2021-01-09.png (c16658f10058...), FinalLightCurve_Qatar-4 b_2021-01-09.csv (196edfb8eb9c...)

Compute resources

Wall time 117.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 01:05Z.